AI Playbook and Toolkit for Public Health Departments

Public Health Interoperability: HL7 v2, CDA, FHIR, and APIs

ARC 220

Public health AI depends on the ability of systems to exchange and interpret data. Interoperability is the discipline that makes this exchange possible across EHRs, laboratories, HIEs, IISs, surveillance systems, vital records systems, and public health platforms. This module introduces the major standards that public health staff encounter, including HL7 v2, CDA, FHIR, implementation guides, APIs, and terminology standards.

Level	applied/foundational
Primary track	Technical Architecture Track
Appears in tracks	technical-architecture

Learning Objectives

- Explain why interoperability is a technical foundation for public health AI.
- Distinguish among HL7 v2 messages, CDA documents, FHIR resources, and APIs.
- Identify where interoperability standards appear in public health workflows.
- Explain the purpose of implementation guides and terminology standards.
- Develop interoperability requirements and vendor questions for AI-supported workflows.

Module Overview

Public health AI depends on the ability of systems to exchange and interpret data. Interoperability is the discipline that makes this exchange possible across EHRs, laboratories, HIEs, IISs, surveillance systems, vital records systems, and public health platforms. This module introduces the major standards that public health staff encounter, including HL7 v2, CDA, FHIR, implementation guides, APIs, and terminology standards.

Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

Interoperability matters because AI tools cannot safely use data they cannot access, parse, or interpret. A public health agency may receive data from hundreds or thousands of healthcare and laboratory partners. Without consistent standards, data exchange becomes a patchwork of custom interfaces, manual uploads, local workarounds, and incomplete fields. AI tools built on top of that environment may appear sophisticated while depending on fragile inputs.

Standards also affect equity and efficiency. When data exchange is inconsistent, some providers or communities may be represented more completely than others. AI-supported surveillance, prioritization, or outreach may then reflect uneven reporting infrastructure rather than true public health need. Interoperability review helps identify these gaps before outputs are used operationally.

For procurement and implementation, interoperability knowledge helps public health staff move beyond broad vendor claims. A vendor statement that a product supports FHIR or APIs is not enough. Staff need to know which resources, profiles, data elements, security controls, terminology mappings, validation methods, and workflows are supported.

Core Concepts

HL7 v2. HL7 v2 is a widely used message-based standard in healthcare and public health. It is common in laboratory reporting, immunization messaging, admission-discharge-transfer notifications, and other interfaces. HL7 v2 is mature and widely deployed, but messages can vary by implementation, making validation and local onboarding important.

CDA. Clinical Document Architecture is a document-based standard used to represent structured clinical documents. CDA has been used in public health reporting contexts, including case reporting. CDA documents can include structured sections and narrative text, which can be useful but also require careful parsing and interpretation.

FHIR. FHIR, or Fast Healthcare Interoperability Resources, represents health information as modular resources and commonly supports API-based exchange. FHIR can make specific data easier to request and reuse, but FHIR compatibility does not guarantee that required public health data are complete or available.

API. An API allows one system to request data or services from another system. In AI workflows, APIs may support retrieval, task creation, status updates, or integration with systems of record. API access must be governed by permissions, authentication, rate limits, logging, and intended use.

Implementation guide. An implementation guide specifies how a standard should be used for a particular purpose. Public health implementation guides help reduce ambiguity by defining required data elements, profiles, value sets, and exchange expectations.

Public Health Example

An immunization information system may receive HL7 v2 VXU messages from providers and pharmacies. If an AI tool is used to identify vaccination gaps or generate reminder lists, it depends on correctly parsed vaccine codes, patient matching, dates, dose validity, and provider information. Interoperability failures can look like epidemiologic gaps even when the problem is actually message quality.

A public health agency exploring AI-supported case summaries from EHR data may need to understand whether the data arrive as CDA documents, FHIR resources, or extracted fields in a surveillance system. Each option affects what the AI can access, how source traceability is preserved, and how errors are corrected. Standards knowledge helps staff evaluate the feasibility and safety of the proposed workflow.

Technical Deep Dive

Interoperability review starts by identifying the sending system, receiving system, exchange standard, data elements, and intended use. Staff should ask whether the workflow uses messages, documents, resources, files, or manual entry. They should also identify required terminology systems, such as LOINC, SNOMED CT, CVX, ICD-10-CM, UCUM, or local codes.

Validation is a central part of interoperability. A message or API response may be syntactically valid but still not fit for a public health use case. For example, a lab result may include a test code but not the specimen source, or an immunization message may include a vaccine code but not enough information to evaluate dose validity. AI readiness requires validation against the intended public health workflow, not only technical message acceptance.

Interoperability requirements should be explicit in AI procurement and implementation. Requirements may include supported standards, required fields, value sets, error handling, data provenance, access controls, audit logs, test scenarios, implementation guide conformance, and documentation. These details help prevent expensive custom workarounds and unsafe manual data movement.

Risks, Failure Modes, and Guardrails

Vague standards claims. Vendors may claim standards support without demonstrating required profiles, data elements, or workflows. Guardrails include conformance testing, test messages, implementation guide references, and acceptance criteria.

Manual workarounds. Weak interoperability can lead to copying and pasting sensitive data into AI tools. Guardrails include approved interfaces, secure environments, and prohibitions on unapproved data movement.

Terminology mismatch. Codes may be missing, local, outdated, or mapped incorrectly. Guardrails include terminology governance, value-set review, and data quality monitoring.

Over-reliance on structured data. Structured exchange may omit context found in notes or local fields. Guardrails include source review, workflow validation, and documentation of known data gaps.

Application to AI-Supported Workflows

Generative AI may summarize structured and narrative data received through interoperable exchange, but it must preserve source traceability. Predictive analytics may depend on standardized fields and terminology, so interoperability gaps can affect model validity. RAG tools may retrieve implementation guides or agency protocols, but document governance is needed to avoid stale guidance. Agentic AI may use APIs to take actions, making permissions, audit logs, and write-back controls essential.

Reflection Questions

Which standards are used in your agency's major data exchange workflows?

Which workflows still depend on manual uploads or copy-paste processes?

What vendor claims would need to be tested before implementation?

Which terminology standards are required for the workflow?

How would interoperability gaps affect AI outputs or equity?

Practical Exercise

Create an interoperability review checklist for one proposed AI-supported workflow. Identify the sending and receiving systems, exchange standard, required data elements, terminology standards, validation tests, error handling, security controls, audit logs, and vendor questions.

Include at least one test scenario that would show whether the exchange supports the public health use case.

Expected Artifact or Evidence

Interoperability checklist

Data exchange diagram

Vendor questions

Test scenario and acceptance criteria

Expected Assignment

- Interoperability checklist
- Data exchange diagram
- Vendor questions
- Test scenario and acceptance criteria

Knowledge Check

- 1. Why does interoperability matter for AI-supported public health workflows?
- 2. What is FHIR commonly used for?
- 3. What is an implementation guide?
- 4. Which question should be asked during AI vendor review?
- 5. What is strong completion evidence for this module?

References and Resources

- HL7 FHIR resources and implementation guides: https://hl7.org/fhir/
- HL7 US Core Implementation Guide: https://build.fhir.org/ig/HL7/US-Core/
- ONC USCDI resources: https://isp.healthit.gov/united-states-core-data-interoperability-uscdi
- CDC Public Health Data Strategy: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- Public health implementation guides for eCR, ELR, IIS, and syndromic surveillance